Supplementary Material for MOSAIC-N: auditable multimodal single-cell annotation across donors and incomplete protein panels

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1 Evidence policy

V33 replaces the random-cell PBMC split as the primary statistical protocol with nested leave-one-donor-out evaluation. Test donor, validation donor and six training donors are disjoint. Gene selection and RNA/ADT transformations are fitted using training donors only. Three model seeds are averaged within each held-out donor, and the eight donors are the biological replicates. Cell counts and seeds are not treated as independent biological observations.

The supplement distinguishes four evidence types: retrained method comparisons, same-checkpoint inference ablations, deterministic missing-panel stress tests and constructed leave-class-out stress tests. PDC101 uses a separate label space and protocol, so it is reported as a compatible direct hierarchy comparison rather than pooled with PBMC. Older random-cell, COVID-19, 5P and COMBAT results are retained as secondary breadth evidence and do not replace the donor-disjoint inference.

2 Dataset lineage and nested splits

The GSE164378 3P and 5P cohorts contain the same eight donors and are not independent-donor datasets. The primary 3P source contains 161,764 cells, 58 L3 labels and 224 proteins. For each fold, 3,000 genes are selected using the six training donors. Every training fold contains all 58 labels, so natural held-out-donor evaluation is closed set.

Table S1: Nested donor split sizes. Validation donors follow a fixed cycle; no donor occurs in more than one partition within a fold.

Test donor	Validation donor	Training cells	Validation cells	Test cells
P1	P2	126,424	17,205	18,135
P2	P3	129,899	14,660	17,205
P3	P4	130,014	17,090	14,660
P4	P5	122,863	21,811	17,090
P5	P6	119,211	20,742	21,811
P6	P7	115,151	25,871	20,742
P7	P8	109,643	26,250	25,871
P8	P1	117,379	18,135	26,250

Split manifests are stored under `configs/splits/pbmc_cite_seq/nested_lodo/`. The aggregate cache audit is `results/tables/mosaic_n_v33_nested_lodo_cache_summary_2026-07-23.csv`. Each cache records label support and confirms that feature selection and scalers were fitted on training donors only.

3 Complete nested-donor performance matrix

Table S2 reports retrained methods and same-checkpoint inference variants. The point estimate for each donor is the mean of seeds 41, 42 and 43. The 95% interval is then calculated across eight donor means. "Worst donor" is the minimum score for performance metrics.

Table S2: Complete nested donor-disjoint performance matrix.

Method	Metric	Mean [95% CI]	Worst donor	n donors
ADT branch	accuracy	0.8360 [0.8245, 0.8474]	0.8149	8
ADT branch	weighted_f1	0.8364 [0.8270, 0.8457]	0.8159	8
ADT branch	macro_f1	0.6661 [0.6515, 0.6808]	0.6360	8
Fusion branch	accuracy	0.9100 [0.9070, 0.9130]	0.9042	8
Fusion branch	weighted_f1	0.9092 [0.9052, 0.9133]	0.9005	8
Fusion branch	macro_f1	0.8059 [0.7919, 0.8198]	0.7771	8
Learned margin gate	accuracy	0.9119 [0.9086, 0.9153]	0.9055	8
Learned margin gate	weighted_f1	0.9107 [0.9062, 0.9153]	0.9010	8
Learned margin gate	macro_f1	0.8077 [0.7925, 0.8228]	0.7794	8
Learned margin gate + HSR	accuracy	0.9119 [0.9086, 0.9153]	0.9057	8
Learned margin gate + HSR	weighted_f1	0.9108 [0.9062, 0.9153]	0.9012	8
Learned margin gate + HSR	macro_f1	0.8072 [0.7917, 0.8227]	0.7792	8
RNA branch	accuracy	0.8894 [0.8846, 0.8942]	0.8827	8
RNA branch	weighted_f1	0.8882 [0.8825, 0.8940]	0.8805	8
RNA branch	macro_f1	0.7564 [0.7402, 0.7726]	0.7293	8
Uniform fusion	accuracy	0.9130 [0.9096, 0.9164]	0.9068	8
Uniform fusion	weighted_f1	0.9118 [0.9072, 0.9165]	0.9021	8
Uniform fusion	macro_f1	0.8114 [0.7961, 0.8266]	0.7803	8
MLP early fusion	accuracy	0.9091 [0.9048, 0.9133]	0.9023	8
MLP early fusion	weighted_f1	0.9068 [0.9020, 0.9116]	0.8987	8
MLP early fusion	macro_f1	0.7923 [0.7758, 0.8089]	0.7626	8
MOSAIC-N full	accuracy	0.9119 [0.9086, 0.9153]	0.9057	8
MOSAIC-N full	weighted_f1	0.9108 [0.9062, 0.9153]	0.9012	8
MOSAIC-N full	macro_f1	0.8072 [0.7917, 0.8227]	0.7792	8
MOSAIC-N without HSR	accuracy	0.9135 [0.9102, 0.9168]	0.9081	8
MOSAIC-N without HSR	weighted_f1	0.9121 [0.9079, 0.9162]	0.9057	8
MOSAIC-N without HSR	macro_f1	0.8097 [0.7950, 0.8243]	0.7847	8
MOSAIC-N without KD	accuracy	0.9134 [0.9102, 0.9167]	0.9092	8
MOSAIC-N without KD	weighted_f1	0.9121 [0.9078, 0.9164]	0.9049	8
MOSAIC-N without KD	macro_f1	0.8110 [0.7956, 0.8265]	0.7813	8

MOSAIC-N full versus MLP produced paired differences of +0.0029 ACC (95% CI −0.0001 to 0.0059; paired $P = 0.0588$), +0.0040 W-F1 (0.0013–0.0066; $P = 0.0101$) and +0.0148 M-F1 (0.0091–0.0206; $P = 0.0005$). Wilcoxon sensitivity P values were 0.0547, 0.0156 and 0.0078, respectively. These unadjusted tests describe three prespecified aggregate metrics and are not generalized to every exploratory per-class comparison.

The direct architecture ablations were unfavorable to the full configuration. Full minus no-HSR was −0.0016 ACC, −0.0013 W-F1 and −0.0025 M-F1. Full minus no-KD was −0.0015, −0.0014 and −0.0038. Same-checkpoint learned margin gate plus HSR minus uniform fusion was −0.0011 ACC, −0.0011 W-F1 and −0.0042 M-F1, with all eight donor differences negative. The full paired table is `results/tables/mosaic_n_v33_paired_donor_statistics_2026-07-23.csv`.

Per-class results exclude donor–class combinations with zero true support rather than assigning F1 zero. The support-aware table is `results/tables/mosaic_n_v33_donor_per_class_summary_2026-07-23.csv`. Predictions into a class absent from a donor's truth set remain penalized in donor-level aggregate metrics.

4 Missing-protein and panel-mismatch stress tests

Twenty-four full checkpoints were evaluated under eight deterministic conditions without retraining or test-label policy selection. Random masks remove the same fixed feature subset for all cells within a fold/seed condition. Marker masks remove six memory-associated or 15 T-cell-associated ADTs after alias normalization.

Table S3: Frozen-checkpoint performance and calibration under protein-panel masks. Intervals use held-out donor after averaging seeds. For ECE, the worst donor is the maximum value.

Scenario	Metric	Mean [95% CI]	Worst donor	n donors
full	weighted_f1	0.9108 [0.9062, 0.9153]	0.9012	8
full	macro_f1	0.8072 [0.7917, 0.8227]	0.7792	8
full	ece	0.0713 [0.0645, 0.0781]	0.0874	8
marker_memory	weighted_f1	0.9082 [0.9038, 0.9126]	0.8991	8
marker_memory	macro_f1	0.8006 [0.7833, 0.8180]	0.7678	8
marker_memory	ece	0.0738 [0.0670, 0.0806]	0.0899	8
marker_tcell	weighted_f1	0.9021 [0.8969, 0.9072]	0.8950	8
marker_tcell	macro_f1	0.7859 [0.7709, 0.8008]	0.7647	8
marker_tcell	ece	0.0748 [0.0664, 0.0832]	0.0934	8
random_10	weighted_f1	0.9074 [0.9015, 0.9132]	0.8957	8
random_10	macro_f1	0.8003 [0.7824, 0.8182]	0.7702	8
random_10	ece	0.0728 [0.0662, 0.0795]	0.0892	8
random_30	weighted_f1	0.9020 [0.8963, 0.9076]	0.8920	8
random_30	macro_f1	0.7904 [0.7742, 0.8066]	0.7634	8
random_30	ece	0.0755 [0.0682, 0.0827]	0.0924	8
random_50	weighted_f1	0.8971 [0.8903, 0.9039]	0.8830	8
random_50	macro_f1	0.7742 [0.7557, 0.7927]	0.7424	8
random_50	ece	0.0764 [0.0669, 0.0859]	0.0985	8
random_70	weighted_f1	0.8921 [0.8855, 0.8988]	0.8813	8
random_70	macro_f1	0.7641 [0.7453, 0.7829]	0.7337	8
random_70	ece	0.0770 [0.0680, 0.0859]	0.0965	8
rna_only	weighted_f1	0.8882 [0.8826, 0.8938]	0.8810	8
rna_only	macro_f1	0.7558 [0.7390, 0.7726]	0.7306	8
rna_only	ece	0.1063 [0.0914, 0.1212]	0.1330	8

The donor-level W-F1 degradation slope over 0/10/30/50/70% random masking was -0.0263 ± 0.0071 per fully missing protein fraction. The M-F1 slope was -0.0624 ± 0.0108. Brier score increased by 0.0411 ± 0.0097 per full missing fraction. ECE slope was less stable ($+0.0081 \pm 0.0072$), whereas the RNA-only ECE increase was clear. These are post hoc robustness measurements of a full-input model, not evidence that the model was trained for arbitrary missing modalities.

5 Leave-class-out rejection and hierarchy fallback

The five target labels were removed from all training and validation labels before each 57-class model was fitted. Unknown test cells retained the removed label only for evaluation. Thresholds were selected from known validation scores at target known coverage 0.95 or 0.80. No unknown validation or test label informed the threshold.

Table S4: Aggregate validation-selected rejection results. Values first average three seeds within each target and then average five leave-class-out targets.

Score	Target coverage	Observed coverage	Unknown recall	AUROC	Parent-safe rate	Hierarchical accuracy
energy	0.80	0.7961	0.4024	0.6066	0.2076	0.9073
energy	0.95	0.9426	0.2881	0.6066	0.1369	0.8967
one_minus_margin	0.80	0.7878	0.4032	0.5948	0.2259	0.9146
one_minus_margin	0.95	0.9409	0.2139	0.5948	0.1140	0.8994
one_minus_max_probability	0.80	0.7887	0.3643	0.5848	0.1981	0.9101
one_minus_max_probability	0.95	0.9414	0.2372	0.5848	0.1231	0.8990

Table S5: Per-target margin-score results at target known coverage 0.80. Support and target difficulty vary substantially.

Target	n unknown	Known coverage	Unknown recall	AUROC	Parent-safe rate	Hierarchical accuracy
B naive lambda	83	0.7927	0.1285	0.4258	0.0723	0.9277
CD4 TCM_1	734	0.7807	0.5722	0.7284	0.2489	0.9167
CD8 TEM_4	1291	0.7897	0.4056	0.5696	0.3945	0.9011
NK_3	614	0.7847	0.3458	0.5474	0.3371	0.9192
gdT_2	583	0.7914	0.5638	0.7026	0.0766	0.9083

B naive lambda is the smallest unknown set ($n = 83$) and produced below-chance AUROC and low recall. CD4 TCM_1 and gdT_2 were easier to rank as unknown, but gdT_2 had low parent-safe fallback. The mean unknown parent-safe rate at the chosen margin operating point was only 0.2259. These results preclude a broad open-set or safe-hierarchy claim.

6 PDC101 comparison with official MMoCHi

PDC101 preprocessing was audited before the final comparison. The h5ad RNA matrix was already log-transformed and the protein matrix was already processed; an initial runner that repeated log and quantile transformations was invalidated and archived separately. The valid runner uses the processed matrices, fits only StandardScaler on training cells, excludes HTO/isotype/control proteins and validates every official external-holdout identifier and truth label.

Table S6: PDC101 same-holdout aggregate comparison. Uncertainty modes are not interchangeable.

Method	Metric	Mean ± uncertainty	Mode
MOSAIC-N	accuracy	0.9077 ± 0.0180	seed Student-t
MOSAIC-N	weighted_f1	0.9068 ± 0.0187	seed Student-t
MOSAIC-N	macro_f1	0.9007 ± 0.0209	seed Student-t
MMoCHi	accuracy	0.9347 ± 0.0110	cell bootstrap; conditional on one workflow
MMoCHi	weighted_f1	0.9335 ± 0.0114	cell bootstrap; conditional on one workflow
MMoCHi	macro_f1	0.9298 ± 0.0118	cell bootstrap; conditional on one workflow

Table S7: PDC101 per-class F1 point estimates on the 2,098-cell sorted holdout.

Class	Support	MOSAIC-N F1	MMoCHi F1
cd4_cm	207	0.8185	0.8435
cd4_em	295	0.9374	0.9420
cd4_n	227	0.9333	0.9358
cd8_cm	223	0.8246	0.8851
cd8_em	365	0.8825	0.9122
cd8_emra	206	0.8863	0.9493
cd8_n	241	0.9260	0.9734
monocyte	334	0.9970	0.9970

MMoCHi exceeded MOSAIC-N for every aggregate metric and most classes. The largest gaps occurred at CD8 central-memory/effector-memory boundaries. Monocyte F1 was identical. MMoCHi intervals are cell bootstrap intervals conditional on one official workflow fit; MOSAIC-N intervals are Student-t margins over three model seeds.

7 Cross-donor feature attribution

Gradient-times-input was aggregated separately for RNA and ADT in six prespecified sibling labels. Donor-pair values first average three seeds. Because the 28 donor pairs overlap, Spearman and Jaccard summaries are descriptive and are not used as 28 independent replicates.

Table S8: Cross-donor attribution stability and canonical-marker enrichment. Marker overlap is the fraction of available canonical markers appearing in the aggregate top 20.

Class	Modality	Mean Spearman	Top-20 Jaccard	Marker overlap	Permutation p
CD4 TCM_3	ADT	0.6123	0.5103	1.0000	0.0002
CD4 TCM_3	RNA	0.7510	0.4355	0.4000	0.0012
CD4 TEM_3	ADT	0.5535	0.4722	0.5000	0.04759
CD4 TEM_3	RNA	0.7400	0.4182	0.5000	0.0002
CD4 TEM_4	ADT	0.2945	0.2915	0.2500	0.3181
CD4 TEM_4	RNA	0.5292	0.1127	0.5000	0.0003999
CD8 Naive	ADT	0.5545	0.4684	0.8000	0.0002
CD8 Naive	RNA	0.7956	0.2895	0.0000	1
CD8 Naive_2	ADT	0.4850	0.4740	0.8000	0.0002
CD8 Naive_2	RNA	0.6829	0.2993	0.2000	0.02999
CD8 TCM_1	ADT	0.5816	0.4810	0.6000	0.006399
CD8 TCM_1	RNA	0.7529	0.4548	0.2000	0.03059

RNA rank stability was highest for CD8 Naive (mean Spearman 0.796) and lowest for CD4 TEM_4 (0.529). ADT stability ranged from 0.295 for CD4 TEM_4 to 0.612 for CD4 TCM_3. Ten of 12 unadjusted marker-permutation tests had $P < 0.05$; CD4 TEM_4 ADT and CD8 Naive RNA did not. This mixed result is retained rather than summarized as universal marker recovery.

Seed-pair mean Spearman values were 0.644–0.843 for RNA and 0.519–0.758 for ADT, generally higher than donor-pair stability. Integrated gradients used seed 42, up to eight correct cells per donor/class and 16 integration steps. Gradient-times-input versus integrated-gradient Spearman was usually above 0.70, but CD4 TEM_4 sometimes had only one or two correct samples. The sanity analysis is therefore directional, not an independent interpretation benchmark.

8 Secondary breadth evidence

V32 analyses are retained for completeness but are not part of the primary nested-donor statistical claim. The fixed 100,000-cell PBMC random split placed every donor in training, validation and test. Ten learned-HSR seeds reached 0.9209 ACC, 0.9200 W-F1 and 0.8437 M-F1, but paired HSR effects crossed zero. This protocol measures within-cohort architecture performance and must not be called unseen-donor generalization.

The E-MTAB-10026 patient-disjoint split assigned 90/20/20 nonoverlapping patients and 453,073/81,318/112,975 cells to training/validation/test. MOSAIC-N versus matched MLP reached 0.8590/0.8550/0.6471 versus 0.8553/0.8514/0.6411 for ACC/W-F1/M-F1. ACC and W-F1 paired intervals were positive; the three-seed M-F1 interval crossed zero. The single-patient B_malignant label was train-only.

GSE164378 5P used the same eight donors as the primary 3P source and a smaller 54-protein panel. A fixed three-seed MOSAIC-N ensemble reached 0.8646/0.8617/0.6916. It is assay-cohort and ensemble evidence, not independent-donor or single-model superiority.

COMBAT/COVID parent-safe experiments compared an attributable MOSAIC-N DCRG adapter, scNym and scANVI on aligned seeds 44/45/46. The separate cross-method consensus reached 0.9788 ACC, 0.9790 W-F1 and 0.8372 M-F1, but it consumes predictions from all three methods and is not a MOSAIC-N single-model result. COMBAT source provenance is recorded in `docs/datasets/combat_citeseq/README.md`.

9 Reproducibility and artifact contract

The formal matrices contain 96 PBMC retraining units, 24 full-checkpoint ablation units, 192 panel conditions, 20 leave-class-out training units and 90 reject operating conditions. PDC101 contains three MOSAIC-N seeds and one official MMoCHi workflow artifact. Each serious training run saves configuration, log, checkpoint, prediction, per-class metrics and artifact index.

V33 table and figure scripts read only complete V33 sources:

- `experiments/exp_baseline_matrix/build_v33_manuscript_tables.py`;
- `experiments/exp_baseline_matrix/plot_v33_evidence_figure.py`;
- `results/tables/mosaic_n_v33_manuscript_evidence_manifest_2026-07-23.json`;
- `results/tables/mosaic_n_v33_evidence_figure_sources_2026-07-23.json`.

The local release builder copies code, configuration, split manifests, small evidence tables, commands and environment metadata and verifies SHA-256 hashes. It excludes raw data, caches and checkpoints. A software license, stable public repository, archived release DOI, final funding statement and confirmed CRediT roles require author or external-account action and are not inferred.

10 Claim boundaries

The evidence supports donor-disjoint W-F1 and M-F1 improvement over matched MLP, graceful but nontrivial degradation under missing proteins, limited validation-selected rejection, a negative compatible comparison with MMoCHi and moderate cross-donor feature-attribution stability. It does not support significant ACC superiority over MLP, aggregate benefit from HSR/KD/learned gating, robustness to arbitrary missing modalities, unrestricted open-set recognition, universal marker recovery, causal feature interpretation or one model dominating all datasets.